

SHETH LUJ AND SIR MV COLLEGE
Subject: Data Analysis with SAS / SPSS / R

Practical No: 15

Aim: Generating basic summaries using str() or summary() (R).

Code:

```
library(dplyr)
```

```
# --- 1. SETUP: Load Data ---
```

```
df <- read.csv("Cancer dataset.csv") %>%  
  select(id, diagnosis, radius_mean, texture_mean, area_mean)
```

```
# 2. USING str() (Structure)
```

```
print("--- OUTPUT OF str() (Data Structure) ---")
```

```
# Shows that 'diagnosis' is a character (chr) and others are numeric (num/int)  
str(df)
```

```
# 3. USING summary() (Statistical Summary)
```

```
print("--- OUTPUT OF summary() (Descriptive Statistics) ---")
```

```
summary(df)
```

Output:

The screenshot shows the RStudio interface. The Source pane contains the following R code:

```
R - R4.5.2 - ~/...  
> library(dplyr)  
> # --- 1. SETUP: Load Data ---  
> df <- read.csv("Cancer dataset.csv") %>%  
+   select(id, diagnosis, radius_mean, texture_mean, area_mean)  
> # 2. USING str() (Structure)  
> print("--- OUTPUT OF str() (Data Structure) ---")  
[1] "--- OUTPUT OF str() (Data Structure) ---"  
> # Shows that 'diagnosis' is a character (chr) and others are numeric (num/int)  
> str(df)  
'data.frame': 569 obs. of 5 variables:  
 $ id      : int  842302 842517 84300903 84348301 84358402 843786 844359 84458202 844981 84501001 ...  
 $ diagnosis: chr  "M" "M" "M" "M" ...  
 $ radius_mean: num  18 20.6 19.7 11.4 20.3 ...  
 $ texture_mean: num  10.4 17.8 21.2 20.4 14.3 ...  
 $ area_mean : num  1001 1326 1203 386 1297 ...  
> # 3. USING summary() (Statistical Summary)  
> print("--- OUTPUT OF summary() (Descriptive Statistics) ---")  
[1] "--- OUTPUT OF summary() (Descriptive Statistics) ---"  
> summary(df)
```

The Console pane shows the output of the str() function:

```
data.frame: 569 obs. of 5 variables:  
 $ id      : int  842302 842517 84300903 84348301 84358402 843786 844359 84458202 844981 84501001 ...  
 $ diagnosis: chr  "M" "M" "M" "M" ...  
 $ radius_mean: num  18 20.6 19.7 11.4 20.3 ...  
 $ texture_mean: num  10.4 17.8 21.2 20.4 14.3 ...  
 $ area_mean : num  1001 1326 1203 386 1297 ...
```

The Environment pane shows the following objects:

Object	Variables
df	569 obs. of 5 variables
df_batch_1	5 obs. of 4 variables
df_batch_2	2 obs. of 4 variables
df_calc	239 obs. of 33 variables
df_clean	239 obs. of 31 variables
df_dates_extrac	569 obs. of 4 variables
df_logic	239 obs. of 33 variables
df_long	5690 obs. of 4 variables
df_subset	569 obs. of 12 variables
df_text	239 obs. of 32 variables
df_wide_recreat	569 obs. of 12 variables
dropped_multiple	239 obs. of 29 variables
dropped_one	239 obs. of 30 variables
dropped_range	239 obs. of 26 variables
duplicate_check	0 obs. of 2 variables
ESGcountry	240 obs. of 31 variables
ESGcountry.Seri	1349 obs. of 4 variables
final_dataset	239 obs. of 34 variables
final_list	10 obs. of 31 variables
merged_data	1349 obs. of 34 variables
omit_target_cols	239 obs. of 5 variables
range_cols	239 obs. of 8 variables
selected_cols	239 obs. of 4 variables
series_data	1349 obs. of 4 variables
split_matrix	chr [1:239, 1:3] "Demographic and Health...
starts_with_1	239 obs. of 7 variables
tidy_data	239 obs. of 38 variables
unique_diagnoses	2 obs. of 1 variable
unique_radius	456 obs. of 3 variables